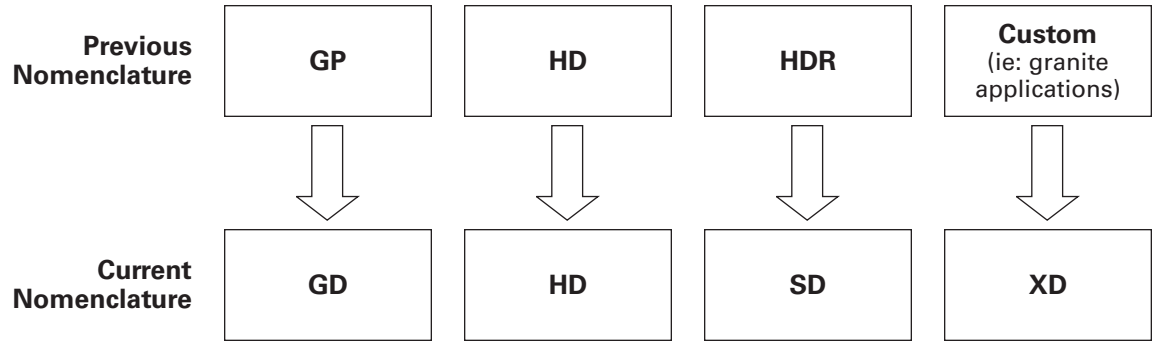
The diagram below illustrates how previous nomenclature aligns with the current, followed by a brief description of each category.



Description:

General Duty
- Low impact
- Lower abrasion materials (ie: dirt, loam, mixed compositions of dirt and fine gravel)
- Pin-on and coupler

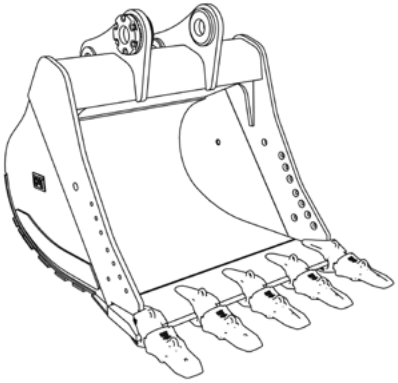
Heavy Duty
- Wide range of impact
- Wide range of abrasion material (ie: mixed dirt, clay and rock)
- Trenching
- Variety of applications
- Pin-on and coupler

Severe Duty
- Higher abrasion conditions (ie: shot granite)
- Wear bars, wear plates are substantially thicker and larger for added protection
- Pin-on and coupler

Extreme Duty
- High abrasion conditions (ie: granite quarries with high quartzite content
- Corner shrouds have been added and side wear plates are larger for added protection against abrasion and gouging wear
- Pin-on and coupler

BUCKET TYPES

General Duty

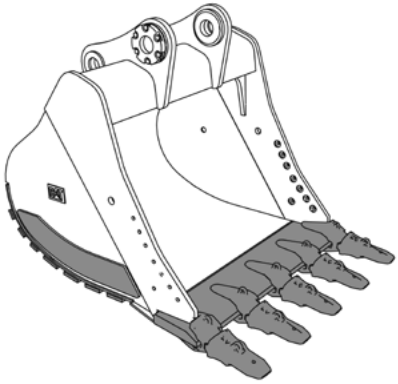


For digging in low impact, lower abrasion materials such as dirt, loam, and mixed compositions of dirt and fine gravel. Example: Digging conditions in which General Duty tip life exceeds 800 hours.

Typically larger General Duty buckets are the most popular sizes, and are used by site developers to mass excavate in low abrasion applications.

- Lighter structures decrease load time and increase the weight that can be lifted.
- Standard size adapters and tips.
- Sidebars are pre-drilled for optional sidecutters and sidebar protectors.

Heavy Duty



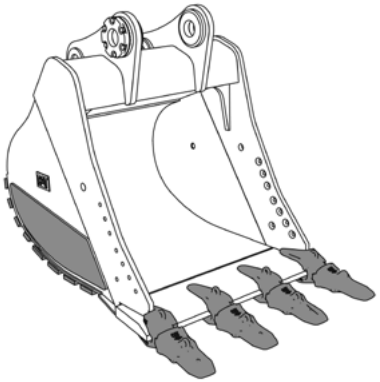
The most popular excavator bucket style. A good "center line" choice, or starting point, when application conditions are not well known.

For a wide range of impact and abrasion conditions including mixed dirt, clay and rock. Example: Digging conditions where Penetration Plus tip life ranges from 400 to 800 hours.

Heavy Duty Buckets are recommended for trenching in utilities work, and for the general contractor working in a variety of different situations.

- Thicker bottom and side wear plates than General Duty buckets for more durability.
- Adapters and tips are up-sized for enhanced performance and durability.
- Sidebars are pre-drilled for optional sidecutters and sidebar protectors.

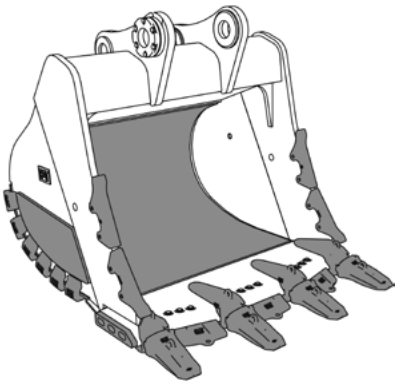
Severe Duty



For higher abrasion conditions such as well shot granite and caliche. Example: Digging conditions where tip life ranges from 200 to 400 hours with Penetration Plus tips.

- Bottom wear plates are about 50% thicker than Heavy Duty buckets.
- Side wear plates are about 40% larger than Heavy Duty buckets for added protection against abrasive and gouging wear.
- Heavy Duty and Severe Duty buckets use same size adapters.
- Adapters are sized to accommodate higher abrasion conditions.
- Tips are up-sized (over the General Duty bucket) for enhanced performance and durability.
- Sidebars are pre-drilled for optional sidecutters and sidebar protectors.

Extreme Duty



For very high abrasion conditions including high quartzite granite. Example: Digging conditions where tip life is less than or equal to 200 hours with Extra Duty tips.

- Corner (or heel) shrouds, Base Edge End Protectors (BEEPs), base edge segments, liners, and Mechanically Attached Wear Plates (MAWPs) protect the bucket from wear.
- Side wear plates are larger.
- Sidebar protection has been added for protection against abrasion and gouging wear.
- Adapters are sized to accommodate higher abrasion conditions.
- Tips are up-sized (over the General Duty bucket) for enhanced performance and durability.

A wide range of specialty buckets to fit your specialty applications. These buckets are designed with specific tasks or applications in mind such as ditch cleaning, grading, and trenching.

Clean-Up

Clean-up buckets combine the straight edge and width options of a Ditch Cleaning bucket with the capacity and durability of a General Duty bucket, creating the ideal bucket for digging, grading, and finishing work.

Ditch Cleaning and Ditch Cleaning Tilt

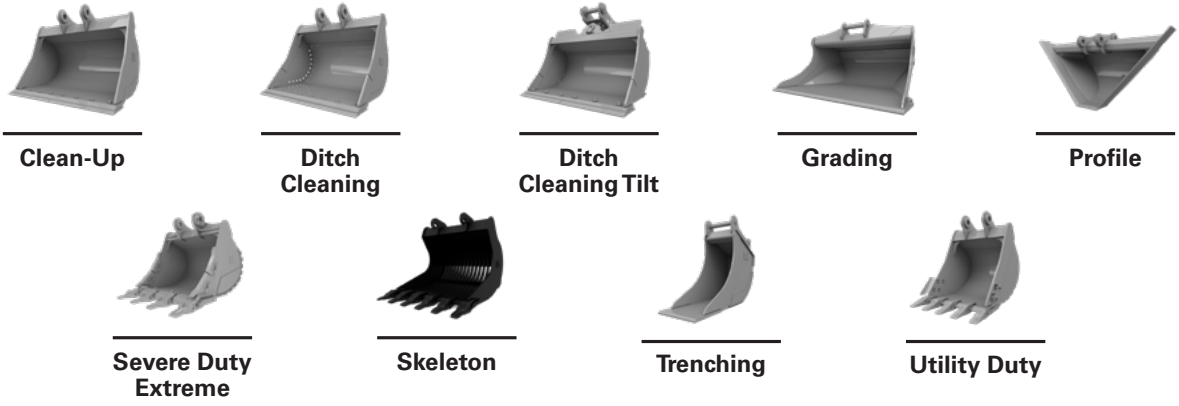
Ditch Cleaning buckets work best for cleaning ditches, sloping, grading, and other finish work as the only style with side liquid drainage holes, making it easier to move solid materials. Ditch Cleaning Tilt buckets have 90° of hydraulic tilting capability, 45° left and 45° right powered by two, double-acting cylinders.

Grading and Trenching

Grading and Trenching buckets are primarily used to create a smooth finish or grade to a job. Primarily used with the Tiltrotator, the bucket gains the capability to create a smooth finish from multiple angles. The narrower profile of Trenching buckets allows for digging specific widths when necessary.

Skeleton

Some applications may require the removal of larger materials while sifting out finer materials, for example removing large rocks from a sandy area. Skeleton buckets are ideal for these situations. Skeleton buckets can excavate, sort, and sift due to their mesh design.



Utility Duty

Utility Duty buckets are a cost-effective solution for your basic digging needs. Ideal applications are low-impact, low-abrasion materials. The shallow profile makes it easier to empty sticky materials such as loam or clay.

Severe Duty Extreme

Severe Duty Extreme buckets are capable of handling, loading, and face scraping of high-impact materials such as high quartzite granite, ore, coal, and shot rock. Designed specifically for 35 ton and higher excavators in highly abrasive applications.

Profile

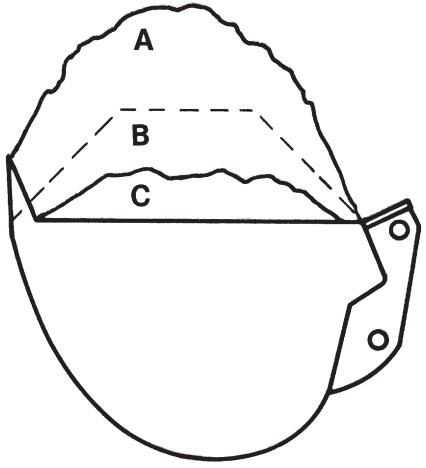
Trench digging is an easy task with the use of a Profile bucket, which has fixed or have adjustable wings, allowing the operator to conform to the size and shape of certain trenches. Profile buckets with wings are adjustable from 30-45° relative to the plane being dug.

BUCKET PAYLOAD

An excavator's bucket payload (actual amount of material in the bucket on each digging cycle) is dependent on bucket size, shape, curl force, and certain soil characteristics, i.e., the fill factor for that soil. Fill factors for several types of material are listed below.

Average Bucket Payload =
(Heaped Bucket Capacity) × (Bucket Fill Factor)

Material	Fill Factor Range (Percent of heaped bucket capacity)
Moist Loam or Sandy Clay	**A** — 100-110%
Sand and Gravel	**B** — 95-110%
Hard, Tough Clay	**C** — 80-90%
Rock — Well Blasted	60-75%
Rock — Poorly Blasted	40-50%



Working Weights — Bucket & Payload

The following tables give maximum "bucket plus payload" weights to assist in selecting the correct bucket for a specific application. These weights are based on actual job conditions. In better than average conditions the excavator may be able to achieve rated lift capacities listed in this section.

NOTE: Bucket sizes are suitable for a maximum material density of 1800 kg/m^3 (3035 lb/yd^3). Payloads shown are calculated at 1500 kg/m^3 (2530 lb/yd^3).

Many Cat excavators have simple to use technologies that help boost operator efficiency that includes Cat Grade with 2D, Cat Grade with Assist, Cat Payload, Cat Lift Assist, and Cat E-Fence. Other available optional upgrades include Cat Grade with Advanced 2D and Cat Grade with 3D.

CAT GRADE WITH 2D

Cat Grade with 2D helps operators reach grade faster. Operators cut and fill to exact specifications without overcutting. It can be programmed to the site's most used target depth and slope offsets so grade can be obtained with ease – a real time saver on the jobsite. No grade checkers are needed so the work area is safer.

CAT GRADE WITH ASSIST

Automated boom, stick, and bucket movements deliver more accurate cuts with less effort. The operator simply sets the depth and slope in the monitor and activates single-lever digging. Also reducing operator fatigue and improving operator accuracy, speed, versatility and safety.

CAT PAYLOAD

Cat Payload technology delivers precise load targets with on-the-go weighing, which helps prevent over/underloading and maximizes efficiency. Automated tracking helps manage production and lower cost. The monitor's USB port lets the operator download the results from production so progress can be managed without needing an internet connection.

LIFT ASSIST

Lift Assist quickly calculates the actual load that is being lifted and compares it to the rated load the excavator is capable of handling. Visual and auditory alerts tell the operators if they are within safe working range or need to take action to avoid tipping.

2D E-FENCE TECHNOLOGY

Whether a bucket or a hammer is being used, standard 2D E-fence automatically stops excavator motion using boundaries operators set in the monitor for the entire working envelope – above, below, sides and front. E-fence protects equipment from damage and reduces fines related to zoning or underground utility damage. Automatic boundaries even help prevent operator fatigue by reducing overswinging or digging.

AVAILABLE OPTIONAL UPGRADES

Cat Grade with Advanced 2D and Cat Grade with 3D increase productivity and expand grading capabilities. Grade with Advanced 2D adds in-field design capabilities through an additional 254 mm (10 in) high-resolution touchscreen monitor. Grade with 3D adds GPS, GNSS, and GLONASS positioning for pinpoint accuracy.

INTRODUCTION

Long reach excavators are designed purposely for light duty digging that requires reach capability well above that of normal digging machines. To be able to have high enough digging forces together with an acceptable size bucket, the long reach excavators have a smaller digging envelope than the ditch cleaning machines. Long reach excavators are ideally suited for deep digging in gravel or sand pits then feeding directly into a hopper.

Caterpillar's long reach hydraulic excavators use purpose-built booms and sticks designed by Caterpillar for maximum performance and durability in light duty applications.

Long Reach Excavation Fronts include: boom, stick, linkage cylinders (boom, stick, and bucket), hydraulic lines, additional counterweight for stability over the side and heavy duty wide undercarriage. Dimensions include light excavation bucket.

INTRODUCTION

Long Reach Excavation machines are designed specifically for jobs requiring longer reach than standard excavators, combined with digging capabilities.

The boom and the stick are purposely designed to perform digging operations with an acceptable bucket size.

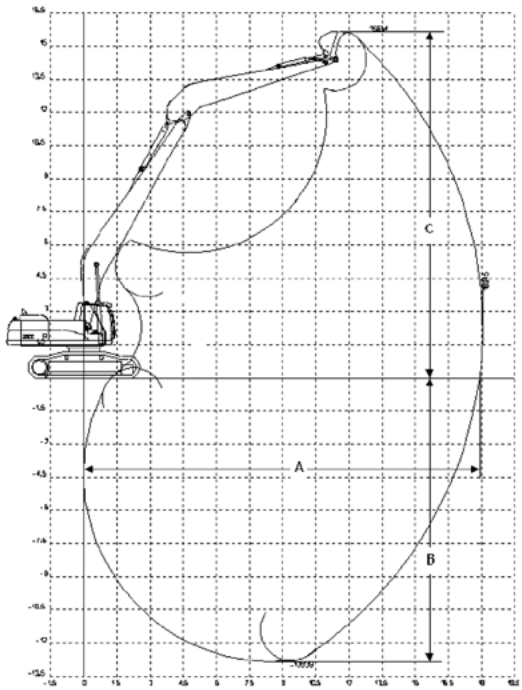
The performances of the machine are attained through the use of bigger boom cylinders, heavy-duty wide undercarriage and significant additional counterweight. A heavy-duty upper-frame is also used in order to guarantee durability and resistance to the extra stresses generated by that demanding application.

Long Reach Excavation machines are ideally suited for deep or long distance digging in sand or gravel pits, slope forming, cleaning of settling banks, river conservation and other work formerly reserved for draglines.

These excavators can of course feed directly into a hopper or load a truck that would stand by their side.

The boom and the stick are designed following Caterpillar's standards, in order to provide the maximum performance and durability in digging applications.

Long Reach Excavation Fronts include: boom, stick, linkage (boom, stick, and bucket cylinders), hydraulic lines and additional counterweight. Dimensions include the bucket.

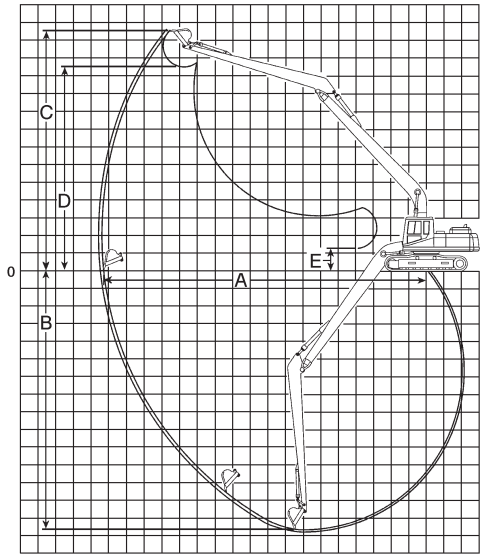


INTRODUCTION

Super long reach excavators are designed specifically for those jobs requiring maximum reach well beyond the range of normal excavators. Those machines are designed to drag a small bucket at about 90 degrees over the side of the tracks towards the excavator; they are not suited for digging work. Caterpillar offers the Long Reach excavators for light digging applications with a much larger digging envelope than normal excavators. Super long reach excavators are suited for ditch cleaning, slope finishing, river conservation and other work formerly reserved to draglines.

Caterpillar's super long reach hydraulic excavators use purpose-built booms and sticks designed by Caterpillar for maximum performance and durability in dragging applications.

Super long reach fronts include: boom, stick, linkage cylinders (boom, stick, and bucket), hydraulic lines and additional counterweight for stability while working over the side. Dimensions include bucket.



MACHINE SELECTION: TRACKS VERSUS WHEELS

Features:

Tracks	Wheels
● Flotation	● Mobility and speed
● Traction	● No pavement damage
● Maneuverability	● Better stability with outriggers or dozers
● Severe underfoot	● Leveling machine with outriggers
● Faster machine repositioning	● Dozing capability

Tracks (10T to 95T)

Unless the application calls for a lot of travel to, from, and around the job sites, a track-type excavator could be the better choice. Tracked-type excavators provide good traction and flotation in almost all kinds of underfoot conditions. Consistently good drawbar power provides excellent maneuverability. The Tracked undercarriage also provides good overall stability. If the job calls for frequent machine repositioning, a track-type excavator will provide better operating efficiency — where raising and lowering outriggers would take extra time.

Wheels (15T to 24T)

Looking for a highly versatile machine? A machine that can do more than mass excavation and trenching. Consider a Wheeled Excavator.

A Wheeled Excavator combines traditional excavator features such as 360° swing, long reach, deep digging depth, high loading height, high digging forces and high lift capacities, with the mobility of a wheeled undercarriage. The rubber tires allow the excavator to travel paved roads, work in shopping malls, squares, parking lots and other paved areas without damaging the pavement. It's mobility allows fast independent travel between job sites as well as on the job site giving you more job planning flexibility. The Wheeled Excavator is the ideal tool for truck loading in tight quarters, undercutting concrete or asphalt, patching, shoulder work, curb and gutter repair, landscaping, spreading top soil, fine grading, laying pipe, placing manholes or ditch cleaning.

A Wheeled Excavator is also an ideal machine in material handling. It can load or unload trucks and carry loads around the job site. Stabilizers and a dozer blade can be pinned to the undercarriage increasing the machine's stability during lifting.

Equip the Wheeled Excavator with dedicated special attachments such as cab riser, material handling stick and boom. Add the additional hydraulic circuit option and your ready for a complete range of special tools. Ditch cleaning bucket, clamshell, grapples, hammers to name a few.

Cat wheeled Excavators offer a load independent, load-sensing, flow distribution hydraulic system that gives the operator absolute precision and control no matter what the application.

Machine weight is the key to selecting a Wheeled Excavator. Following are some additional factors that need to be considered.

Choose the proper boom and stick for your reach, digging depth and lifting requirements. Stability can be greatly enhanced by adding outriggers and/or a dozer blade. Additional hydraulic circuits can be added depending on your application and stick end attachments.

Acceptable Bucket/Stick Combinations

The following charts identify the acceptable bucket and stick combinations for Cat wheeled Excavators and are based on stability. Minimum stability occurs with the linkage oriented over the side and positioned as shown in the visual. Dozer and/or stabilizers (if equipped) are raised and the bucket contains a full load. The longest stick is shown that has acceptable stability for each bucket. That stability is 1.1 moment ratio or better. Once this stability factor is established, all shorter sticks are then acceptable with the listed bucket.

EXCAVATOR SHOE SELECTION

Undercarriage life can be extended by equipping the machine properly for the application.

Many excavators work on pavement or flat, soft ground and experience few undercarriage problems. But if those same machines (usually equipped with wide track pads) were placed in severe underfoot conditions, undercarriage destruction could occur very rapidly.

The rule, used for other track-type machines — *"Whenever possible use the narrowest shoes available"* — is even more valid for excavators.

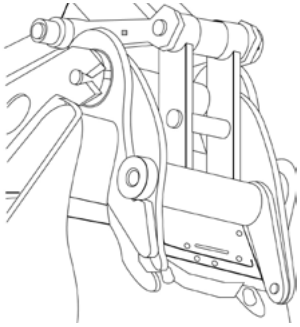
The best general purpose track shoe is the triple grouser. It has a good section modulus and offers the best compromise between traction and minimum disturbance to paved surface.

The double grouser shoe has a better section modulus and is more aggressive than the triple grouser section. Single grouser shoes are offered for maximum traction. Some users like single grousers for added mobility in hilly terrain.

QUICK COUPLER SYSTEMS

Quick couplers can greatly increase a machine's versatility and productivity. They make it much easier to switch attachments which can increase utilization. Quick couplers also encourage changing buckets when the application changes, rather than continue to use a less efficient bucket. Example: An application that is predominately dirt with occasional pockets or seams of rock. Without a quick coupler the owner may choose to live with a rock bucket but, rock buckets are normally smaller and heavier which reduces performance in a dirt application. A quick coupler allows the use of the rock bucket in the rock and a GP bucket in the dirt.

Caterpillar offers two main types of quick couplers. The first is a dedicated type. A typical system substitutes hooks on the bucket for the pin-on hinges used with conventional buckets. The mating portion is pinned on the stick and bucket linkage. It slips into the hooks to secure the bucket or other attachment.



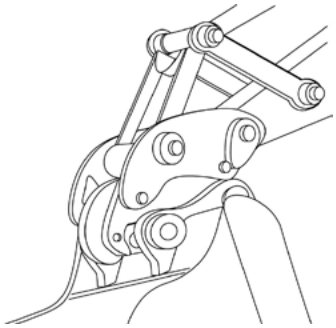
Advantages:

Bucket tip radius (distance from the bucket pivot point to the bucket tip) does not increase. Increased tip radius reduces curl and stick forces which can reduce the loadability of the bucket. The hook type coupler also does not add appreciable weight at the end of the stick. Keeping the tip radius and weight the same ensure no compromise in performance. The machine portion of the hook type coupler can be designed to allow more than one machine to use the same buckets.

Disadvantages:

The hook type system requires special buckets. Conventional pin-on buckets cannot be used. The ability to use buckets on more than one machine requires careful application analysis. Larger machines generate forces that can destroy the wrong buckets. Smaller machines with the wrong bucket may develop loads in excess of the machine's capability. Even if the machine can handle the loads, the tip radius may be too large to allow the bucket to load properly. With the flexibility of a quick coupler comes the responsibility to make sure the bucket or other attachments are properly sized for each application.

The second type of quick coupler is the pin grabber type. This device pins on the stick and bucket linkage and grabs the bucket pins on standard pin-on buckets.



Advantages:

The advantage of the pin grabber is that it will pick up standard pin-on buckets. No need to purchase new attachments that will fit the system.

Disadvantages:

The pin grabber is mounted between the stick and the bucket which increases the tip radius. The amount of increase depends on the pin grabber's manufacturer. Increasing tip radius can compromise performance by decreasing bucket forces. The coupler also adds weight and reduces the payload capability.

Pin grabbers are required to mate up with existing bucket pins. Because different machines require different pin spreads and diameters, they offer very limited ability to match with buckets from other machines.

Both types of quick couplers offer cab activated… this allows an attachment change in 30 seconds or less.

Pin grabber couplers allow work tools to be changed quickly — improving overall production and increasing machine versatility. A coupler is pinned on in place of the bucket with standard pins, and can be easily removed should the need arise to mount a tool directly to the stick.

The Pin Grabber Coupler is designed for use on machines sized from 10T to 95T — and is designed to engage and disengage the same range of work tools as previous pin grabber couplers.

No dimensional or interface changes have been made to this coupler. However, a new concept in the locking mechanism inside Center-Lock brings many benefits to the operator.

Compliant

The Cat Pin Grabber Coupler gives operators confidence through its locking system and visible locking mechanism. Couplers meet or exceed latest EN and ISO safety standards: EN474 and ISO/DIS 13031.

Visible Secondary Lock

The Pin Grabber Coupler was designed with the operator in mind. The secondary lock is clearly visible from the cab, providing an obvious indicator of coupler status: open or closed. The ability to see the lock on the front pin give the operator confidence and makes changing tools faster, while giving everyone on the job site reassurance that the Pin Grabber is locked tight.

Job Site Confidence

From engagement, all while working, to the time the attachment is disengaged, you can be completely confident attachments are connected properly.

From the operator's seat, visual and audible indicators help assure attachment is coupled. Your Cat excavator hydraulics, mechanisms inside the coupler, along with digging forces assure the attachment operates as expected.

The Cat Pin Grabber Coupler meets and exceeds safety standards allowing you to work anywhere in the world.

Productivity

As with any quick coupler, the Pin Grabber changes between attachments in seconds, allowing one machine to be used for multiple tasks on the job site.

The ease of operation and quickness of tool changeover maximizes productivity on the job site. The coupler is easy to operate, regardless of user skill level, and it's easier to train new operators to use. The ability to see when the coupler is open and closed saves time whenever a tool is changed. The elimination of the locking bar allows for operation regardless of boom, stick and machine position.

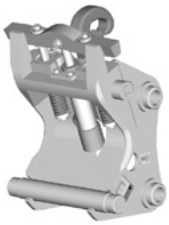
The Pin Grabber Coupler can pick up many buckets in reverse "front shovel" position for better control in utility work and precise digging and grading operations. Many competitive buckets can be engaged, making the Cat Pin Grabber invaluable in mixed fleets or rental fleets.

The Cat CW quick coupler can pick up any work tool and is equipped with a wedge-style locking system minimizing break-out force loss. The CW is highly suitable for harsh applications, such as demolition and quarries. The CW has become the industry standard in Europe. It is interchangeable with different machine classes, and has been designed for use with more than 700 different machines, both Cat and non Cat.

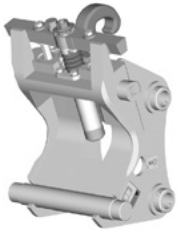
The Cat CW Series is available in a spindle and hydraulic version. A spindle version can easily be modified into a hydraulic version and vice versa.

Additional Benefit:

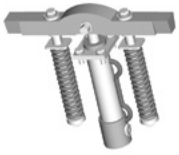
Hoisting hooks — To make the CW Series more versatile, hoisting hooks are available from 2 metric tons (2.2 tons) up to 20 metric tons (22 tons) capacity for maximum lifting capacity.

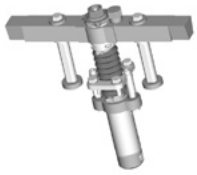


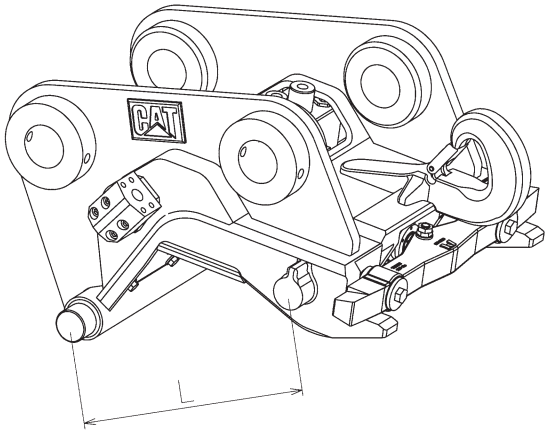
Hydraulic Version

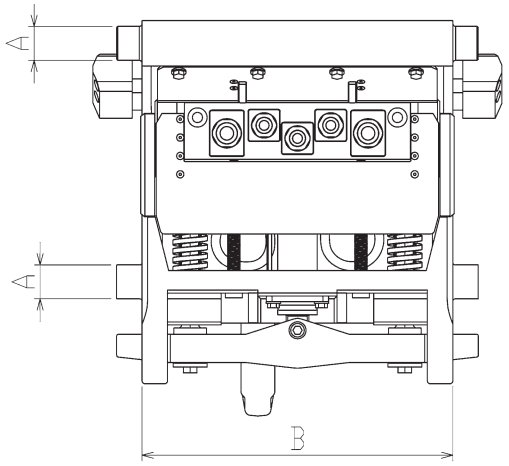


Spindle Version









Auto-Connect Quick Coupler

The Auto-Connect quick coupler automates tool exchange fully, so operators can change work tools quickly, from the safety and comfort of their cabs. Tool changes become a matter of seconds. Built on the field proven quick coupler CW platform, the unique design of the Auto-Connect prevents ruptured hoses and oil spills, avoiding unplanned downtime.

Full Contamination Control on Both Quick Coupler and Work Tool:

A sliding cover keeps any dust and debris away from the hydraulic area, protecting the fluid connectors when not in use. Seals on both the Quick Coupler and the work tool cover have been included to close the gap with the cartridges, ensuring full contamination control. Rigid work tools without hydraulics, like buckets, do not need conversion to be picked up by the Auto-Connect quick coupler, avoiding unnecessary expenses. Due to the location of the hydraulic coupling unit, it's protected against damage from outside.

Plug-and-Perform System:

Auto-Connect is a "plug-and-perform" system based on Caterpillar's dedicated CW platform, and fits directly to the machine and existing controls. The Auto-Connect is controlled via the Quick coupler actuation circuit. Combined with Cat Tool Control, it's really easy for operators to select tools and make fast changeovers. With features like Auto Depressurization as standard no additional hydraulic components are required.

RIPPING & LOADING IN QUARRIES

The "Rip & Load" concept includes a large mass excavator equipped with a hydraulic quick coupler, a rock bucket and a ripper tine. The ripper tine is used to disrupt the in-situ rock formation, after which same excavator switches to the bucket to load the rock. This system is used where economical, environmental or legal issues prevent or restrict the use of explosives. In these situations, depending on geology, ripping either reduces the amount of explosives necessary, or replaces explosives all together.

Advantages:

- Reduction or elimination of blasting costs.
- Reduced safety risks.
- Smaller environmental impact (less noise and vibrations).
- Less exposure to precipitation, resulting in less water damage.
- Less waste (up to 35% reduction).
- Less internal cracks, resulting in higher quality product.
- Work areas can be closer to existing infrastructures.
- Fewer machines and personnel.
- Increased versatility with quick coupler (different buckets, hammers).
- Lower Cost per Ton.

Rip and Load Hourly Production

(With Hydraulic Quick Coupler)

Model	Metric Tons/Hour	Short Tons/Hour
45T	150 - 300	**165 - 330**
65T-74T	200 - 400	**220 - 440**
85T-95T	300 - 500	**330 - 550**

Rippability

Refer to "Tip Selection", "Estimating Ripping Production" and "Use of Seismic Velocity Charts" in the Track-type Tractors section. This information generally applies to usage of a ripper tine on the mass excavator.

Rippability Comparison between LHEX and LTTT

The excavator ripping technique is different from production ripping with a track-type tractor. The track-type tractor pulls the ripper(s) through the rock mass at a constant rate, whereas the excavator uses its stick- and curl forces to break material away from a horizontal or vertical face. Forward visibility in the excavator allows the operator to position the ripper tooth and attack geological discontinuities to assist the ripping process.

In ripping and loading, the ripper is typically used between 15% and 20% of the hour preparing the material. Tool change time, when using the hydraulic quick coupler, is insignificant with 2% to 6%. The remainder of the time is used for loading.

The ripping process improves bucket penetration which will increase service life of the bucket.

CYCLE TIME ESTIMATING CHARTS

The digging cycle of the excavator is composed of four segments:

1. Load Bucket
2. Swing Loaded
3. Dump Bucket
4. Swing Empty

Total excavator cycle time is dependent on machine size (small machines can cycle faster than large machines) and job conditions. With excellent job conditions the excavator can cycle fast. As job conditions become more severe (tougher digging, deeper trench, more obstacles, etc.), the excavator slows down accordingly. As the soil gets harder to dig, it takes longer to fill the bucket. As the trench gets deeper and the spoil pile larger, the bucket has to travel farther and the upper structure has to swing farther on each digging cycle.

Spoil pile or truck location also affects cycle time. If a truck is located on the floor of the excavation beside material being moved, 10 to 17 second cycles are practical. The other extreme would be a truck or spoil pile located above the excavator 180° from the excavation.

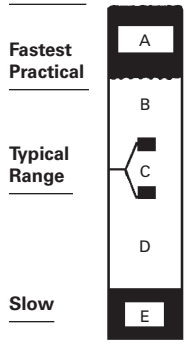
In sewer construction work the operator may not be able to work at full speed because he has to dig around existing utilities, load the bucket inside a trench shield, or avoid people working in the area.

The Cycle Time Estimating Chart outlines the range of total cycle time that can be expected as job conditions range from excellent to severe. Many variables affect how fast the excavator is able to work. The chart defines the range of cycle times frequently experienced with a machine and provides a guide to what is an "easy" or a "hard" job. The estimator can then evaluate the conditions of his job and use the Cycle Time Estimating Chart to select the appropriate working range. A practical method of further calibrating the Cycle Time Estimating Chart is to observe excavators working in the field and correlate measured cycle times to job conditions, operator ability, etc.

The following table breaks down what experience has shown to be typical Cat excavator cycle times with:

— no obstruction in the right of way
— above average job conditions
— an operator of average ability and
— 60°-90° swing angle.

These times would decrease as job conditions or operator ability improved and would get slower as conditions become less favorable.



KEY

A — Excellent
B — Above Average
C — Average
D — Below Average
E — Severe

CYCLE TIME -vs- JOB CONDITION DESCRIPTION

— Easy digging (unpacked earth, sand gravel, ditch cleaning, etc.). Digging to less than 40% of machine's maximum depth capability. Swing angle less than 30°. Dump onto spoil pile or truck in excavation. No obstructions. Good operator.

— Medium digging (packed earth, tough dry clay, soil with less than 25% rock content). Depth to 50% of machine's maximum capability. Swing angle to 60°. Large dump target. Few obstructions.

— Medium to hard digging (hard packed soil with up to 50% rock content). Depth to 70% of machine's maximum capability. Swing angle to 90°. Loading trucks with truck spotted close to excavator.

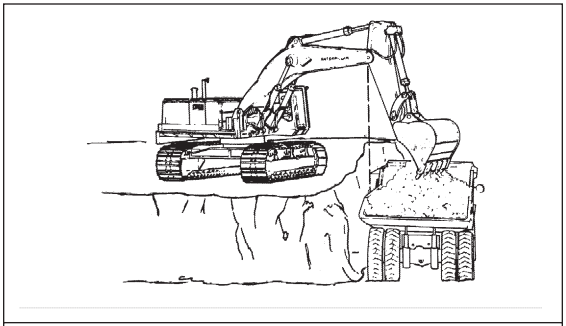
— Hard digging (shot rock or tough soil with up to 75% rock content). Depth to 90% of machine's maximum capability. Swing angle to 120°. Shored trench. Small dump target. Working over pipe crew.

— Toughest digging (sandstone, caliche, shale, certain limestones, hard frost). Over 90% of machine's maximum depth capability. Swing over 120°. Loading bucket in man box. Dump into small target requiring maximum excavator reach. People and obstructions in the work area.

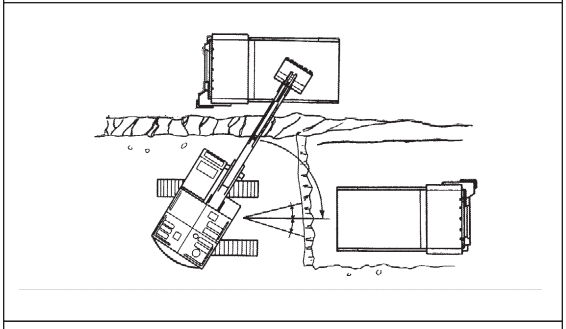
CYCLE TIME ESTIMATING CHART												
CYCLE TIME	Tracked 10T-11T	Tracked 12T Wheeled 14T-15T	Tracked 15T Wheeled 16T-18T	Tracked 19T Wheeled 20T	Tracked 20T Wheeled 22T	Tracked 23-24T	Tracked 30T	Tracked 36T	Tracked 49T	Tracked 65-74T	Tracked 85T-95T	CYCLE TIME
10 sec.												**0.17** min.
15												**0.25** min.
20 sec.												**0.33** min.
25												**0.42** min.
30 sec.												**0.50** min.
35												**0.58** min.
40 sec.												**0.67** min.
45												**0.75** min.
50 sec.												**0.83** min.
55												**0.92** min.
60 sec.												**1.0** min.

Cat 300 Series Mass Excavation booms and buckets coupled with the proper stick will help you move material faster and more efficiently in production excavation and loading applications. With the largest bucket, shortest stick and long undercarriage your excavator can often do the work of a larger machine. A longer stick and standard undercarriage make it ideal for loading on-highway trucks and general construction jobs.

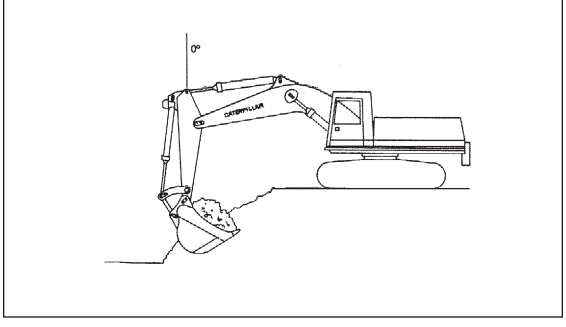
MAXIMIZING PRODUCTION WITH A MASS EXCAVATOR



Ideal Bench Height and Truck Distance — For stable or consolidated materials, bench height should be about equal to stick length. For unstable materials it should be less. The most useful truck position is when the inside truck body rail is below the boomstick hinge pin.



Optimum Work Zone and Swing Angle — For maximum production, the work zone should be limited to 15° either side of machine center or about equal to undercarriage width. Trucks should be positioned as close as possible to machine centerline. Two alternatives shown here.



Best Distance from the Edge — The machine should be positioned so that the stick is vertical when the bucket reaches full load. If the unit is farther back, breakout force is reduced. If it is closer to the edge, undercutting may occur and time is wasted bringing the stick back out. Also, the operator should begin boom-up when the bucket is 75% of the way through the curl cycle. This should be as the stick nears the vertical position.

This example reflects the ideal situation. Not all points are usable on each job, but incorporation of as many of these points as possible will positively affect production.

SELECTING A MASS EXCAVATOR

Selecting a mass excavator model for optimum production requires matching the machine and bucket to the customer's production requirements, material, and haulers. The following 6-Step selection process will help you to consider the key factors which will impact machine selection. Failure to consider these key elements in the selection process may result in choosing a machine that is too small to efficiently handle the desired bucket size or to meet the production requirement. Selecting a mass excavator which is too large may lead to excessive loader wait time, creating excessive "load shocks" into the hauler, and/or overloading the hauler capacity.

Step 1

Determine the material type and bucket fill factor

Refer to the bucket fill factors table.

Example:

Average Blasted Rock = 75 to 90%

Step 2

Estimate the Cycle Time

Refer to the cycle time estimating chart.

Example:

$$\frac{\text{Tracks 65T Excavator in Hard Rock Digging}}{\text{Shot Rock}} = .43 \text{ to } .52 \text{ minute}$$

Step 3

Calculate the Effective Cycles per Hour

Divide the 60 minute hour by cycle time and adjust for availability and efficiencies.

Example:

Cycle Time	0.48 minute
$\frac{\textit{60 minute hour}}{\textit{Cycle Time}}$	$\frac{60}{0.48} = 125$
Operator Skill/Efficiency	0.9 (90%)
Machine Availability	0.95 (95%)
Gen Operational Efficiency	0.83 (50 min/hr)
Effective Cycles per Hour	125 × .9 × .95 × .83 = 89

Step 4

Calculate the Required Bucket Capacity

Divide hourly production requirement by effective cycles per hour, adjust for material density and fill factor.

Example (Metric):

Hourly Production Required	500 Tons/hour
Effective cycles/hour	89
$\frac{\textit{Hourly Production Required}}{\textit{Effective cycles/hour}} = \textit{Required Payload}$	$\frac{500}{89} = 5.6$
Material Density/Loose	1.6 Ton/m^3
$\frac{\textit{Required Payload}}{\textit{Material Density/Loose}} = \textit{Bucket Payload Volume}$	$\frac{5.6}{1.6} = 3.5\ m^3$
Fill Factor	0.85 (85%)
$\frac{\textit{Bucket Payload Volume}}{\textit{Fill Factor}} = \textit{Nominal Bucket Size}$	$\frac{3.5}{.85} = 4.1\ m^3$

Example (English):

Hourly Production Required	550 tons/hour
Effective cycles/hour	89
$\frac{\textit{Hourly Production Required}}{\textit{Effective cycles/hour}} = \textit{Required Payload}$	$\frac{550 \times 2000}{89} = 12{,}360$ lb
Material Density/Loose	2700 lb/yd^3
$\frac{\textit{Required Payload}}{\textit{Material Density/Loose}} = \textit{Bucket Payload Volume}$	$\frac{12{,}360}{2700} = 4.6\ yd^3$
Fill Factor	0.85 (85%)
$\frac{\textit{Bucket Payload Volume}}{\textit{Fill Factor}} = \textit{Nominal Bucket Size}$	$\frac{4.6}{.85} = 5.4\ yd^3$

Step 5

Select Mass Excavator for required bucket size

Refer to machine Technical Specifications to compare models and bucket ranges. Confirm bucket type, size and maximum material density in Technical Specifications for desired model.

Example:

Required bucket capacity approximately
4.1 m^3 (5.4 yd^3)

45T Track Excavator ME bucket capacity
to 3.5 m^3 (4.6 yd^3)

65T Track Excavator ME bucket capacity
to 5.3 m^3 (6.9 yd^3)

85T Track Excavator ME bucket capacity
to 5.6 m^3 (7.3 yd^3)

Best Choice 65T Track Excavator ME
with 4.0 m^3 (5.2 yd^3)
Rock Bucket rated to 1.8 Ton/m^3 (3000 lb/yd^3)
material density in Technical Specification

Important: Re-calculate Steps 2 - 5 based on cycle times of model selected.

Step 6

Select Haulers

General rule for matching trucks is based on number of cycles to fill the truck.

ME:	4 to 6 passes
Front Shovels:	3 to 5 passes

Example (Metric):

Bucket Selected	4 m^3
Volume in 5 passes	5 × 4 × .85 = 17 m^3
Payload	17 × 1.6 = 27.2 Tons
Consider weight of Liners	27.2 + 2 = 29.2 Tons

Suitable Truck Match Options:

35T AT with capacity	19.2 m^3/31.8 t
37T-40T OHT with capacity	24.2 m^3/37.9 t

Example (English):

Bucket Selected	5.2 yd^3
Volume in 5 passes	5 × 5.2 × .85 = 22.1 yd^3
Payload	22.1 × 2700 = 59,670 lb
Consider weight of Liners	59,670 + 4400 lb = 64,070 lb

Suitable Truck Match Options:

35T AT with capacity	25.1 yd^3/70,000 lb
37T-40T OHT with capacity	31.7 yd^3/83,570 lb

EARTHMOVING PRODUCTION

As with any other piece of material handling equipment, excavator earthmoving production is dependent on average bucket payload, average cycle time and job efficiency. If an estimator can accurately predict excavator cycle time and bucket payload, a machine's earthmoving production can be derived from the following formula.

$$\text{m}^3\ (\text{yd}^3)/60\ \text{min hr} = \text{Cycles}/60\ \text{min hr} \times \text{Avg. Bucket Payload in m}^3\ (\text{yd}^3)$$

$$\text{m}^3\ (\text{yd}^3)/60\ \text{min hr} = \frac{60\ \text{min/hr}}{\text{Cycle Time – min}} \times \text{Avg. Bucket Payload in m}^3\ (\text{yd}^3)$$

$$\text{Avg. Bucket Payload} = \text{Heaped Bucket Capacity} \times \text{Bucket Fill Factor}$$

$$\text{Actual m}^3\ (\text{yd}^3)/\text{hr} = \text{m}^3\ (\text{yd}^3)/60\ \text{min hr} \times \text{Job Efficiency Factor}$$

The Production Estimating Tables (next page) will provide theoretical earthmoving production in cubic meters (yards) per hour if bucket size and cycle time can be estimated. The use of an average cycle time allows adjusting the estimated production for specific job sites and applications. For instance, estimating truck loading applications should include truck exchange times which extends the average cycle time and reduces production potential. The values in the table are based on a 60 minute work hour or 100% efficiency (a condition that is never achieved in reality). The estimator should apply a job efficiency factor to the figures in the table based on his judgment or knowledge of actual job conditions.

Areas outlined on the Production Estimating Table define the work ranges of excavators in the size classes of Cat 10T to 95T Excavators. The upper limit on each area corresponds to the "fastest practical" cycle time for the machines. The width of each area corresponds to the range of bucket payload sizes the machine can handle. An unshaded box has been provided in each machine area to provide a guide indicating that the upper limit of earthmoving production is being approached. When working beyond the values in the white area, the estimator should be certain that excellent job conditions will be encountered (easy digging, shallow trench, good operator, etc.).

The Production Estimating Table can also serve as a guide when selecting the proper size machine to do a job, as is shown in the following example.

Example problem (Metric)

Contractor has a job to move 15 300 Bm^3 (19 100 Lm^3 considering 25% swell factor) of wet sandy loam material in rear dump on-highway trucks which will be loaded by an excavator. Average face depth will be 2.4 m with 60-90 degree average swing angle. Ten days are available to do the work. Contractor plans to work 10 hrs/day and estimates a 50 min. work hour (83% job efficiency). He has two excavators that could be made available to do the work — a 320 with 1.0 m^3 bucket or a 336 with 1.9 m^3 bucket. Experience has shown that either machine can get its rated capacity in the sandy loam soil. Could this job be done with either machine or will the 336 have to be used?

Solution: The excavator must produce 1900 Lm^3/ Day (19 100 Lm^3 ÷ 10 Days) which means the required average hourly rate will be 190 Lm^3/60 Min. Hr. (1900 Lm^3/ Day ÷ 10 hrs/day). Further considering the 83% job efficiency, the excavator's capability will have to be 230 Lm^3/ 50 min hr.

The production estimating table shows that the Track 20T excavator with a 1.0 m^3 bucket would have to achieve a 17.1 sec. average cycle time to produce the required 190 Lm^3/ 60 min. hr. With job efficiency applied a 15.0 second average cycle time is required to produce the 230 Lm^3/ 50 min. hr. The 336 with a 1.9 m^3 bucket could obtain the same 60 min. hr. production level with a 35 second average cycle, or 30 second cycles to meet the 50 min. hr. production requirement. The cycle times estimating chart shows that the Track 20T excavator would be working near its maximum capability to meet the production requirement, whereas, the Track 36T excavator could handle the job easily. This information can then be weighed against what else is known about the job (reach requirements, job conditions, operator ability, etc.) to decide whether or not the larger machine is needed.

Example problem (English)

Substitute these English values in the preceding problem:
Job — 20,000 BCY (25,000 LCY considering 25% swell).
Average face depth — 8-12 ft
Track 20T excavator with 1.25 yd^3 bucket or Track 36T excavator with 2.5 yd^3 bucket.

Solution: The excavator must produce 2500 LCY/ Day, which means the required average hourly rate will be 250 LCY/60 min hr. Further considering the 83% job efficiency the excavator's capability will have to be 300 LCY/50 min hr.

The same concluding comments regarding the Production Estimating Table apply here as in the Metric example.

Cubic Meters per 60 Minute Hour*

ESTIMATED CYCLE TIMES		ESTIMATED BUCKET PAYLOAD** — LOOSE CUBIC METERS																			ESTIMATED CYCLE TIMES	
Cycle Time																						
Seconds	Min.	0.2	0.3	0.5	0.7	0.9	1.1	1.3	1.5	1.7	1.9	2.1	2.3	2.5	2.7	2.9	3.1	3.3	3.5	4.0	Cycles Per Min.	Cycles Per Hr.
10.0	0.17																				6.0	360
11.0	0.18																				5.5	330
12.0	0.20	60	90	150	210	270															5.0	300
13.3	0.22	54	81	135	189	243	297	351	405	459	513	567	621	675	729	783	837	891	945	1080	4.5	270
15.0	0.25	48	72	120	168	216	264	312	360	408	456	504	552	600	648	696	744	792	840	960	4.0	240
17.1	0.29	42	63	105	147	189	231	273	315	357	399	441	483	525	567	609	651	693	735	840	3.5	210
20.0	0.33	36	54	90	126	162	198	234	270	306	342	378	414	450	486	522	558	544	630	720	3.0	180
24.0	0.40	30	45	75	105	135	165	195	225	255	285	315	345	375	405	435	465	495	525	600	2.5	150
30.0	0.50	24	36	60	84	108	132	156	180	204	228	252	276	300	324	348	372	396	420	480	2.0	120
35.0	0.58	20	31	51	71	92	112	133	153	173	194	214	235	255	275	296	316	337	357	408	1.7	102
40.0	0.67					81	99	177	135	153	171	189	207	225	243	261	279	297	315	360	1.5	90
45.0	0.75									133	148	164	179	195	211	226	242	257	273	312	1.3	78
50.0	0.83																				1.2	72

Cubic Yards per 60 Minute Hour*

ESTIMATED CYCLE TIMES		ESTIMATED BUCKET PAYLOAD** — LOOSE CUBIC YARDS																			ESTIMATED CYCLE TIMES	
Cycle Time																						
Seconds	Min.	0.25	0.50	0.75	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	3.75	4.00	4.50	5.00	5.25	Cycles Per Min.	Cycles Per Hr.
10.0	0.17																				6.0	360
11.0	0.18																				5.5	330
12.0	0.20	75	150	225	300	375															5.0	300
13.3	0.22	67	135	202	270	337	404	472	540	607	675	742	810	877	945	1012	1080	1215	1350	1417	4.5	270
15.0	0.25	60	120	180	240	300	360	420	480	540	600	660	720	780	840	900	960	1080	1200	1260	4.0	240
17.1	0.29	52	105	157	210	262	315	367	420	472	525	577	630	682	735	787	840	945	1050	1102	3.5	210
20.0	0.33	45	90	135	180	225	270	315	360	405	450	495	540	585	630	675	720	810	900	945	3.0	180
24.0	0.40	37	75	112	150	187	225	262	300	337	375	412	450	487	525	562	600	675	750	787	2.5	150
30.0	0.50	30	60	90	120	150	180	210	240	270	300	330	360	390	420	450	480	510	600	630	2.0	120
35.0	0.58	36	51	77	102	128	154	180	205	231	256	282	308	333	360	385	410	462	513	535	1.7	102
40.0	0.67					112	135	157	180	202	225	247	270	292	315	337	360	405	450	472	1.5	90
45.0	0.75									180	200	220	240	260	280	300	320	360	400	409	1.3	78
50.0	0.83																				1.2	72

Job Efficiency Estimator

Work Time/Hour	Efficiency
60 Min	100%
55	91%
50	83%
45	75%
40	67%

*Actual hourly production = (60 min. hr. production) × (Job Efficiency Factor)
**Estimated Bucket Payload = (Amount of Material in the Bucket)
= (Heaped Bucket Capacity) × (Bucket Fill Factor)
Unshaded area indicates average production.

Cubic Meters/Yards per 60 Minute Hour*

ESTIMATED CYCLE TIMES		ESTIMATED BUCKET PAYLOAD** — LOOSE CUBIC METERS/YARDS											ESTIMATED CYCLE TIMES	
Cycle Time													Cycles Per	Cycles Per
Seconds	Min.	5.0	6.0	7.0	8.0	9.0	10.0	11.0	12.0	13.0	14.0	15.0	Min.	Hr.
15.0	0.25	1200	1440	1680	1920	2160	2400	2640	2880	3120	3360	3600	4.0	240
17.1	0.29	1050	1260	1470	1680	1890	2100	2310	2520	2730	2940	3150	3.5	210
20.0	0.33	900	1080	1260	1440	1620	1800	1980	2160	2340	2520	2700	3.0	180
24.0	0.40	750	900	1050	1200	1350	1500	1650	1800	1950	2100	2250	2.5	150
30.0	0.50	600	720	840	960	1080	1200	1320	1440	1560	1680	1800	2.0	120
35.0	0.58	510	612	714	816	918	1020	1122	1224	1326	1428	1530	1.7	102
40.0	0.67	450	540	630	720	810	900	990	1080	1170	1260	1350	1.5	90
45.0	0.75	390	468	546	624	702	780	858	936	1014	1092	1170	1.3	78
50.0	0.83	360	432	504	576	648	720	792	864	936	1008	1080	1.2	72
55.0	0.92	330	396	462	528	594	660	726	792	858	924	990	1.1	66
60.0	1.00	300	360	420	480	540	600	660	720	780	840	900	1.0	60

Cubic Meters/Yards per 60 Minute Hour*

ESTIMATED CYCLE TIMES		ESTIMATED BUCKET PAYLOAD** — LOOSE CUBIC METERS/YARDS										ESTIMATED CYCLE TIMES	
Cycle Time												Cycles Per	Cycles Per
Seconds	Min.	16.0	17.0	18.0	19.0	20.0	21.0	22.0	23.0	24.0	25.0	Min.	Hr.
15.0	0.25	3840	4080	4320	4560	4800	5040	5280	5520	5760	6000	4.0	240
17.1	0.29	3360	3570	3780	3990	4200	4410	4620	4830	5040	5250	3.5	210
20.0	0.33	2880	3060	3240	3420	3600	3780	3960	4140	4320	4500	3.0	180
24.0	0.40	2400	2550	2700	2850	3000	3150	3300	3450	3600	3750	2.5	150
30.0	0.50	1920	2040	2160	2280	2400	2520	2640	2760	2880	3000	2.0	120
35.0	0.58	1632	1734	1836	1938	2040	2142	2244	2346	2448	2550	1.7	102
40.0	0.67	1440	1530	1620	1710	1800	1890	1980	2070	2160	2250	1.5	90
45.0	0.75	1248	1326	1404	1482	1560	1638	1716	1794	1872	1950	1.3	78
50.0	0.83	1152	1224	1296	1368	1440	1512	1584	1656	1728	1800	1.2	72
55.0	0.92	1056	1122	1188	1254	1320	1386	1452	1518	1584	1650	1.1	66
60.0	1.00	960	1020	1080	1140	1200	1260	1320	1380	1440	1500	1.0	60

Job Efficiency Estimator

Work Time/Hour	Efficiency
60 Min	100%
55	91%
50	83%
45	75%
40	67%

*Actual hourly production = (60 min. hr. production) × (Job Efficiency Factor)
**Estimated Bucket Payload = (Amount of Material in the Bucket)
= (Heaped Bucket Capacity) × (Bucket Fill Factor)

NOTE: For estimating truck loading production include approximately 0.7 minutes for truck exchange time.